Hector V3.2.0 Tutorial

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Hector is an open-source, object-oriented, Reduced Complexity Earth Systems Model (RCM). It models critical earth system processes on a global scale. As a computationally efficient model, Hector has many applications. It can be used to explore scenarios, generate large ensembles of results, explore key uncertainties, and emulate Earth System Models. There are three ways to use Hector: as a stand-alone executable, through the online HectorUI, and as an R package. Installation instructions and examples are available at https://github.com/jgcri/hector and https://github.com/jgcri/hectorui.

Follow the appropriate installation for your machine to run locally. Here you will learn how to use the R hector package to run complete a Hector run, change a model parameter, and query results.

Getting started

```
library(hector)
library(ggplot2)
theme_set(theme_bw(base_size = 14))
options(epr.plot.height=8)
```

```
files <- list.files(system.file("input", package = "hector"))
cat(files, sep= "\n")
```

```
## hector_ssp119.ini
## hector_ssp126.ini
## hector_ssp245.ini
## hector_ssp370.ini
## hector_ssp434.ini
## hector_ssp460.ini
## hector_ssp534-over.ini
## hector_ssp585.ini
## tables
```

Run SSP2-4.5

```
ini_file = system.file("input/hector_ssp245.ini", package = "hector")
hcore = newcore(inifile = ini_file, name = "SSP2-45")
run(core = hcore)
```

```
## Hector core: SSP2-45
## Start date:  1745
## End date:    2300
## Current date:    2300
## Input file:  /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/hector/input/hect
```

```r
vars = c(GLOBAL_TAS(), CONCENTRATIONS_CO2())
cat(vars, sep = "\n")
```

```
## global_tas
## CO2_concentration
```

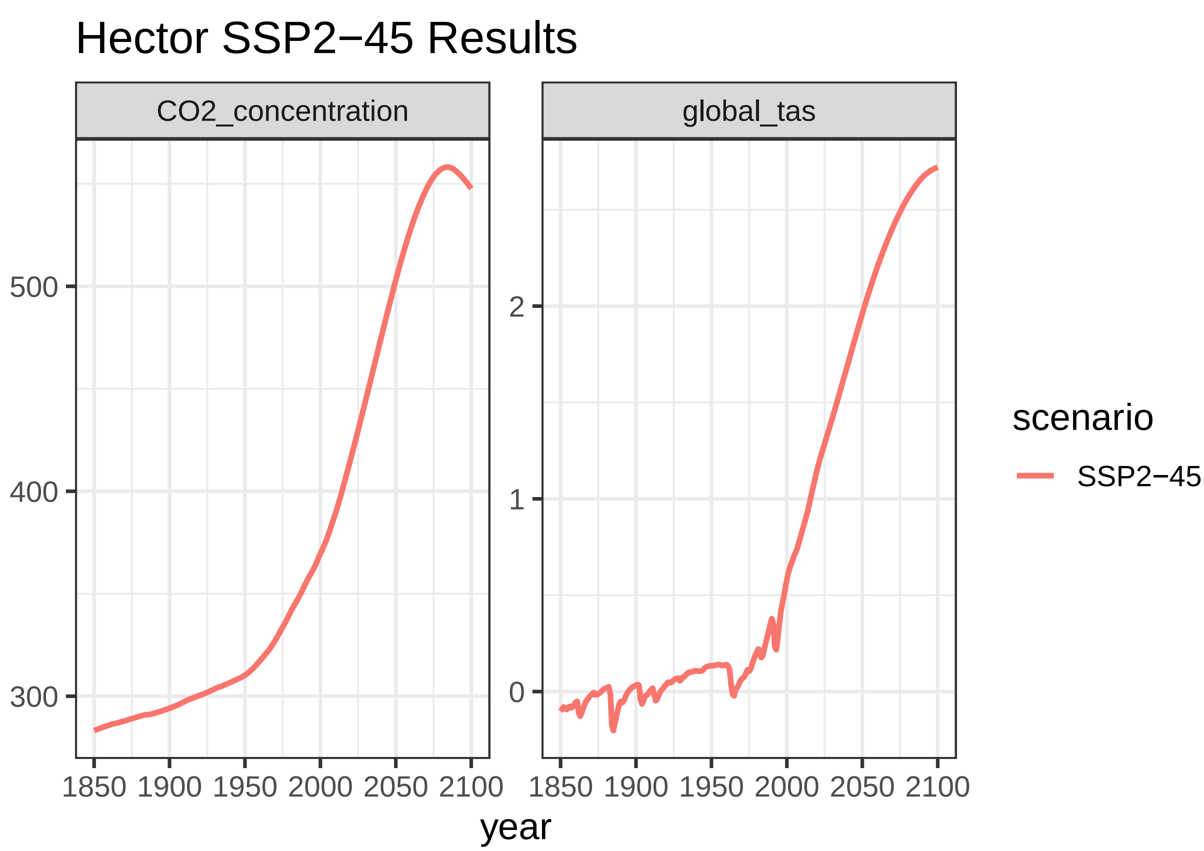
```r
out_ssp245 = fetchvars(core = hcore, dates = 1850:2100, vars = vars)
head(out_ssp245)
```

```
##   scenario year   variable       value units
## 1  SSP2-45 1850 global_tas -0.10249176  degC
## 2  SSP2-45 1851 global_tas -0.09068423  degC
## 3  SSP2-45 1852 global_tas -0.08036054  degC
## 4  SSP2-45 1853 global_tas -0.08557470  degC
## 5  SSP2-45 1854 global_tas -0.09308445  degC
## 6  SSP2-45 1855 global_tas -0.08554412  degC
```

```r
shutdown(hcore)
```

```
## Hector core (INACTIVE)
```

```r
ggplot(data = out_ssp245) +
    geom_line(aes(year, value, color = scenario), linewidth = 1) +
    labs(title = "Hector SSP2-45 Results") +
    facet_wrap("variable", scales = "free") +
    labs(y = NULL)
```



How to change a Hector parameter

```r
ini_file = system.file("input/hector_ssp245.ini", package = "hector")
hcore = newcore(inifile = ini_file, name = "SSP2-45 higher ECS")
run(core = hcore)
```

```
## Hector core: SSP2-45 higher ECS
## Start date:  1745
## End date:    2300
## Current date:    2300
## Input file:  /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/hector/input/hecto
```

```r
default_ECS = fetchvars(core = hcore, dates = NA, vars = ECS())
default_ECS
```

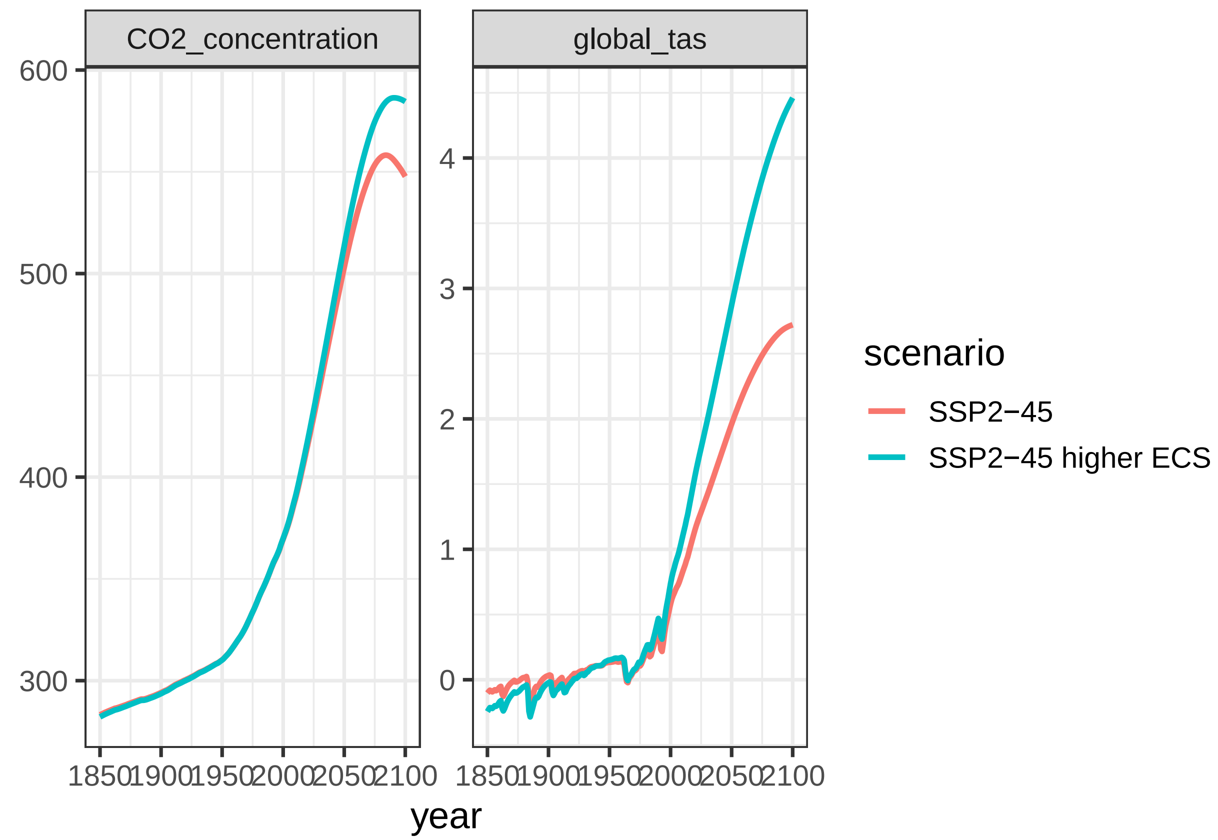
```
##             scenario year variable value units
## 1 SSP2-45 higher ECS   NA        S     3  degC
```

```r
double_ECS = 2 * default_ECS$value
ECS_units = getunits(vars = ECS())

setvar(core = hcore,
       dates = NA,
       var = ECS(),
       values = double_ECS,
       unit = ECS_units)

run(core = hcore)
```

```
## Auto-resetting core to 0
```

```
## Hector core: SSP2-45 higher ECS
## Start date:  1745
## End date:    2300
## Current date:    2300
## Input file:  /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/hector/input/hecto
```

```r
out_ssp245_newECS = fetchvars(core = hcore, dates = 1850:2100, vars = vars)
results <- rbind(out_ssp245, out_ssp245_newECS)

ggplot(data = results) +
    geom_line(aes(year, value, color = scenario), linewidth = 1) +
    facet_wrap("variable", scales = "free") +
    labs(y = NULL)
```



Modifying an emissions pathway

```
ini_file = system.file("input/hector_ssp245.ini", package = "hector")
hcore = newcore(inifile = ini_file, name = "SSP2-45")
run(core = hcore)
```

```
## Hector core: SSP2-45
## Start date:  1745
## End date:    2300
## Current date:    2300
## Input file:  /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/hector/input/hecto
```

```
original_results <- fetchvars(core = hcore,
                              dates = 1850:2100,
                              vars = c(FFI_EMISSIONS(), GLOBAL_TAS()))
original_results$scenario <- "Original SSP2-45"
```

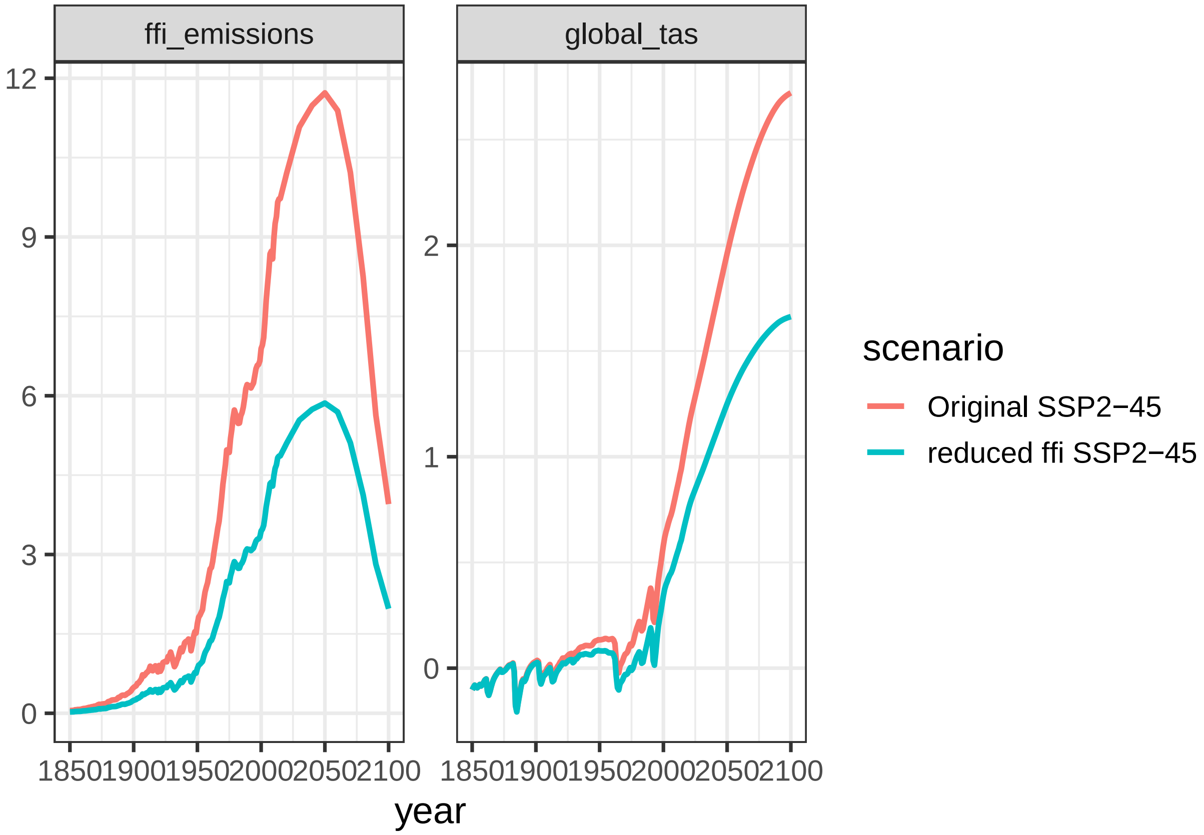
```
ffi_emissions <- fetchvars(core = hcore,
                           dates = 1850:2100,
                           vars = FFI_EMISSIONS())
ffi_emissions$value <- ffi_emissions$value * 0.5
```

```
setvar(core = hcore,
       dates = ffi_emissions$year,
       var = ffi_emissions$variable,
       values = ffi_emissions$value,
       unit = ffi_emissions$units)
```

```r
run(hcore)
```

```
## Auto-resetting core to 1849

## Hector core: SSP2-45
## Start date:  1745
## End date:    2300
## Current date:    2300
## Input file:  /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/hector/input/hecto
```

```r
reduced_ffi_results <- fetchvars(core = hcore,
                            dates = 1850:2100,
                            vars = c(FFI_EMISSIONS(), GLOBAL_TAS()))
reduced_ffi_results$scenario <- "reduced ffi SSP2-45"

results <- rbind(original_results, reduced_ffi_results)

ggplot(data = results) +
    geom_line(aes(year, value, color = scenario), linewidth = 1) +
    facet_wrap("variable", scales = "free") +
    labs(y = NULL)
```



More examples: http://jgcri.github.io/hector/